Machine Learning

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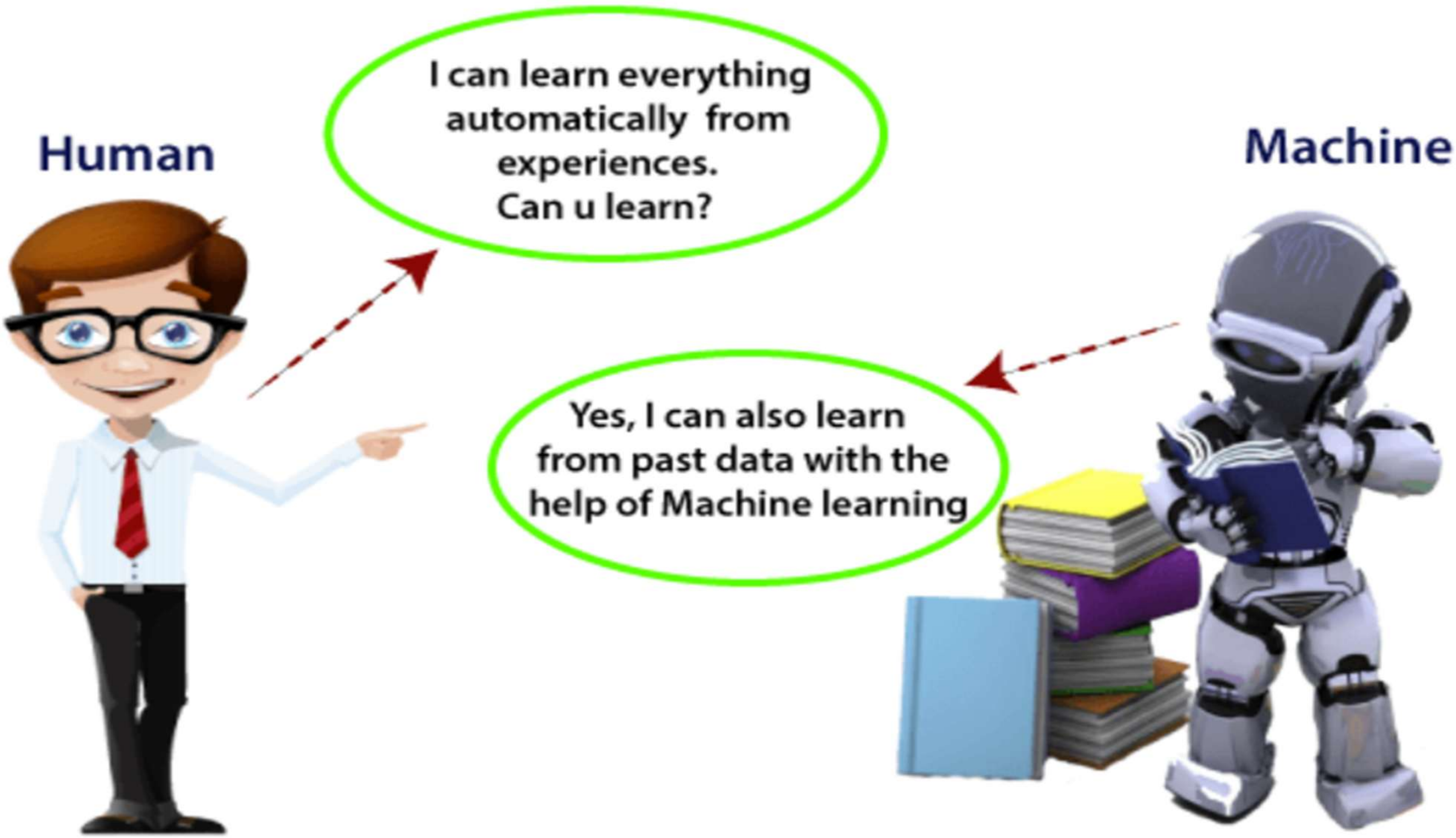
Overview

- Course Overview
- Fundamental of Machine Learning

Machine Learning?

- Informal Definition:
  - Algorithms that can learn from observational data and can make predictions based on it.
  - Machine learning is the practice of computer programming that uses algorithms to learn from data and make intelligent predictions without being explicitly programmed.
  - Machine learning is a growing technology which enables computers to learn automatically from past data.
- That’s pretty much what the human’s brain actually does too.


Human
I can learn everything automatically from experiences. Can u learn?
Machine
Yes, I can also learn from past data with the help of Machine learning

The early age of ML

- 1834 (Charles babbage), the father of modern computer, conceived a device that could be programmed with punch cards, and logical structure.
- 1936 (Alan Turning), designed a theory that how a machine can determine and execute a set of instructions.
- 1940 (ENIAC), first electronic general-purpose computer
- 1950 (Alan Turing), published a paper “Computer Machinery and Intelligence”, on the topic of AI. He asked, Can machine think?

Machine Learning (History)

- Machine Learning is said as a subset of **artificial intelligence.**
- AI as we know it has been officially founded in 1956 at Dartmouth College, where the most eminent experts gathered to brainstorm on intelligence simulation.
- The term machine learning was first introduced by **Arthur Samuel** in **1959**.

ML Should be:

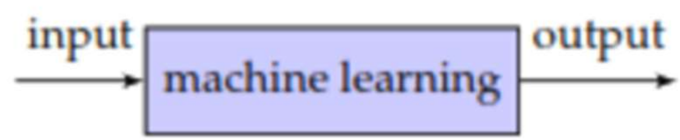
- **A machine has the ability to learn if it can improve its performance by gaining more data.**

Machine Learning

- Formal Definition:
  - Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed. (Arthur Samuel, 59)
  - A computer program is said to learn from experience **E** with respect to some task **T** and some performance measure P, if its performance on **T**, as measured by **P**, improves with experience **E**. (Tom Mitchell, 97)

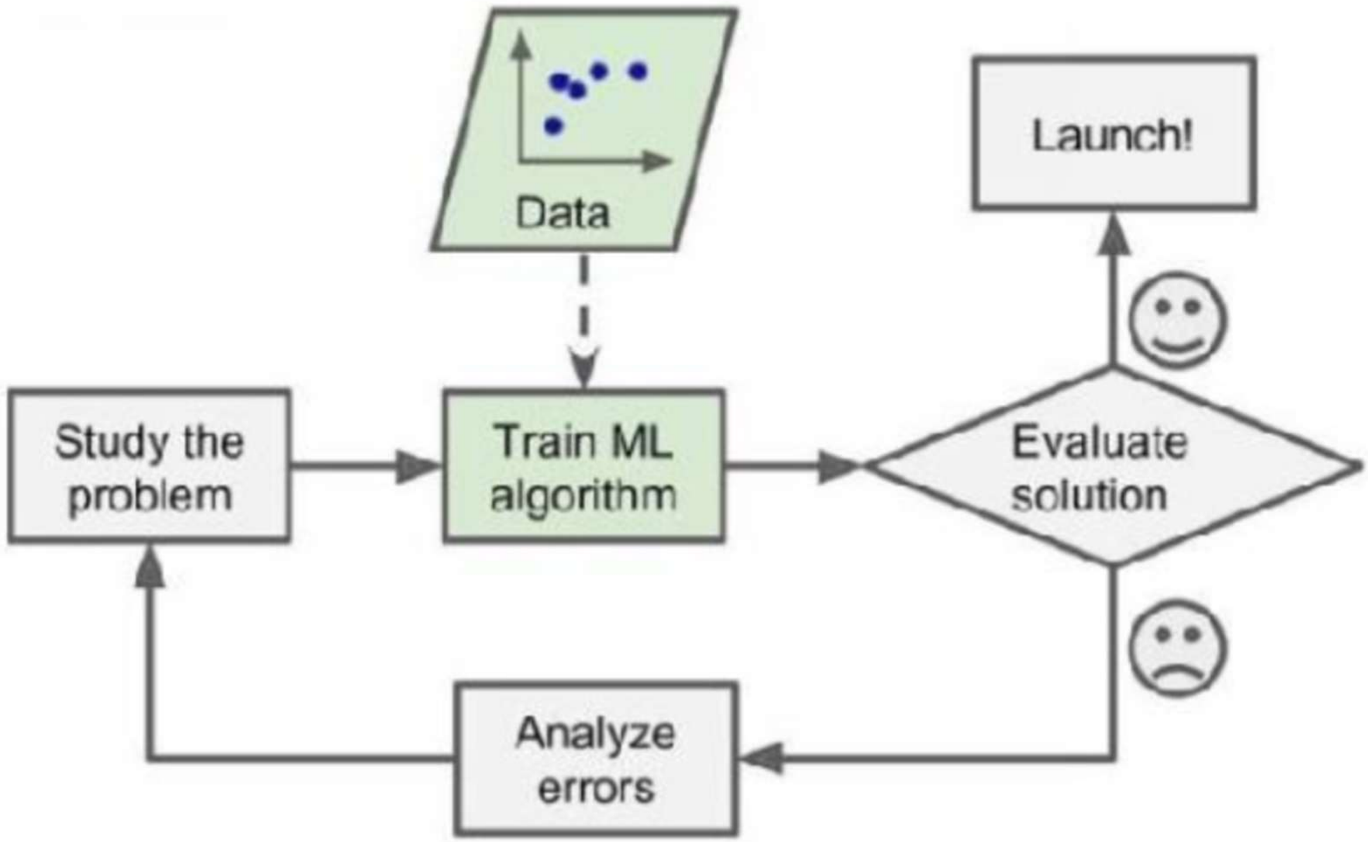
Mathematical Notation of Formal Def. 1:

- f(X)= algorithm
- f(X)→ Y (output)
- f(x1, x2, x3)→ y
  - x1,x2,x3 are features



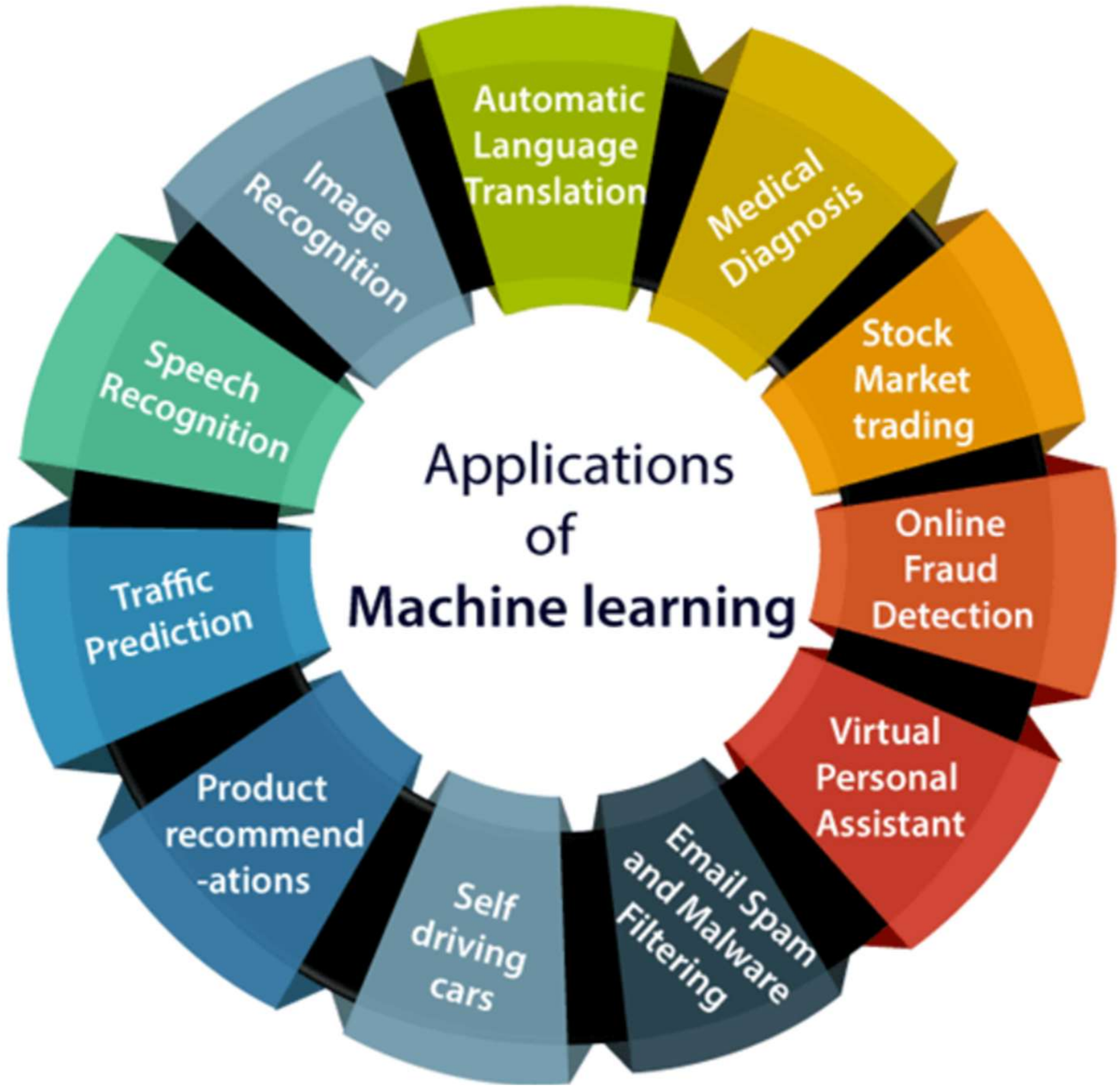
How it works (ML is iterative process)?

- Study the problem
- Train the algorithm
- Evaluate the solution
- If it is good, launch it
- Else, analyze the errors



Mathematical Notation of Formal Def. 2:

- Email Spam Detection System
  - T= 0 or 1 (Spam yes or no)
  - E= Experience (learned from previous cases)
  - P= Performance (Prediction accuracy)
- f(X)=algorithm
- Evaluate (check the performance)
- Launch the system (if performance is good)
- Error analysis (go to f(x) and hyperparameter or check)


Applications
of
Machine learning
Automatic
Language
Translation
Medical
Diagnosis
Stock
Market
trading
Online
Fraud
Detection
Virtual
Personal
Assistant
Email Spam
and Malware
Filtering
Self
driving
cars
Product
recommend
-ations
Traffic
Prediction
Speech
Recognition
Image
Recognition

Types of ML

- We can classify machine learning systems according to the type and amount of human supervision during the training. You can find four major categories:
  - Supervised learning
  - Unsupervised learning
  - Semi-supervised learning
  - Reinforcement learning
  - Self-supervised learning

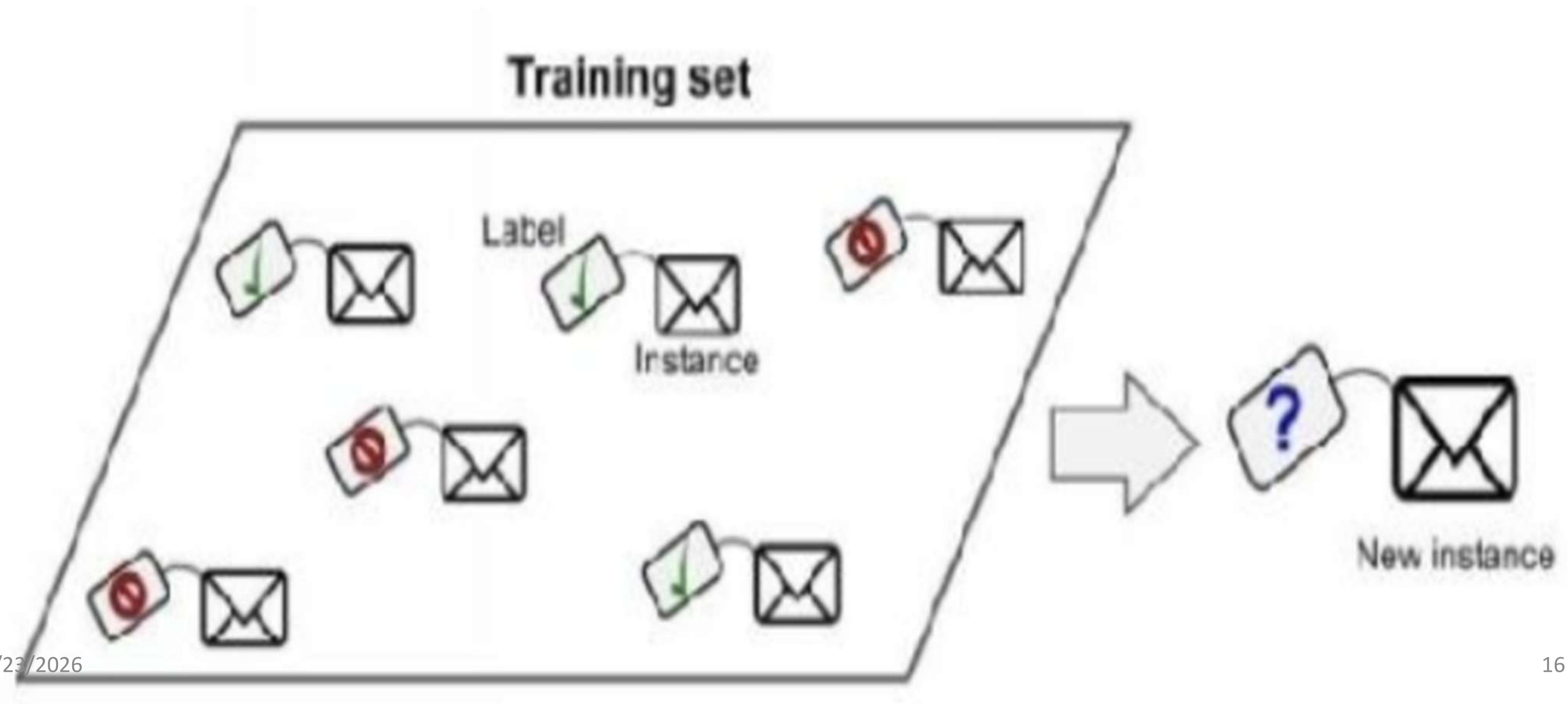
1. Supervised learning

- In this, we provide sample data, with the desired output (labels) to the machine in order to train it, and on that basis, it predicts the unseen output.
- The training data in supervised learning consist of input–output pairs. For each input in the training data, we know its corresponding output, which can be used to guide learning algorithms as a supervision signal.

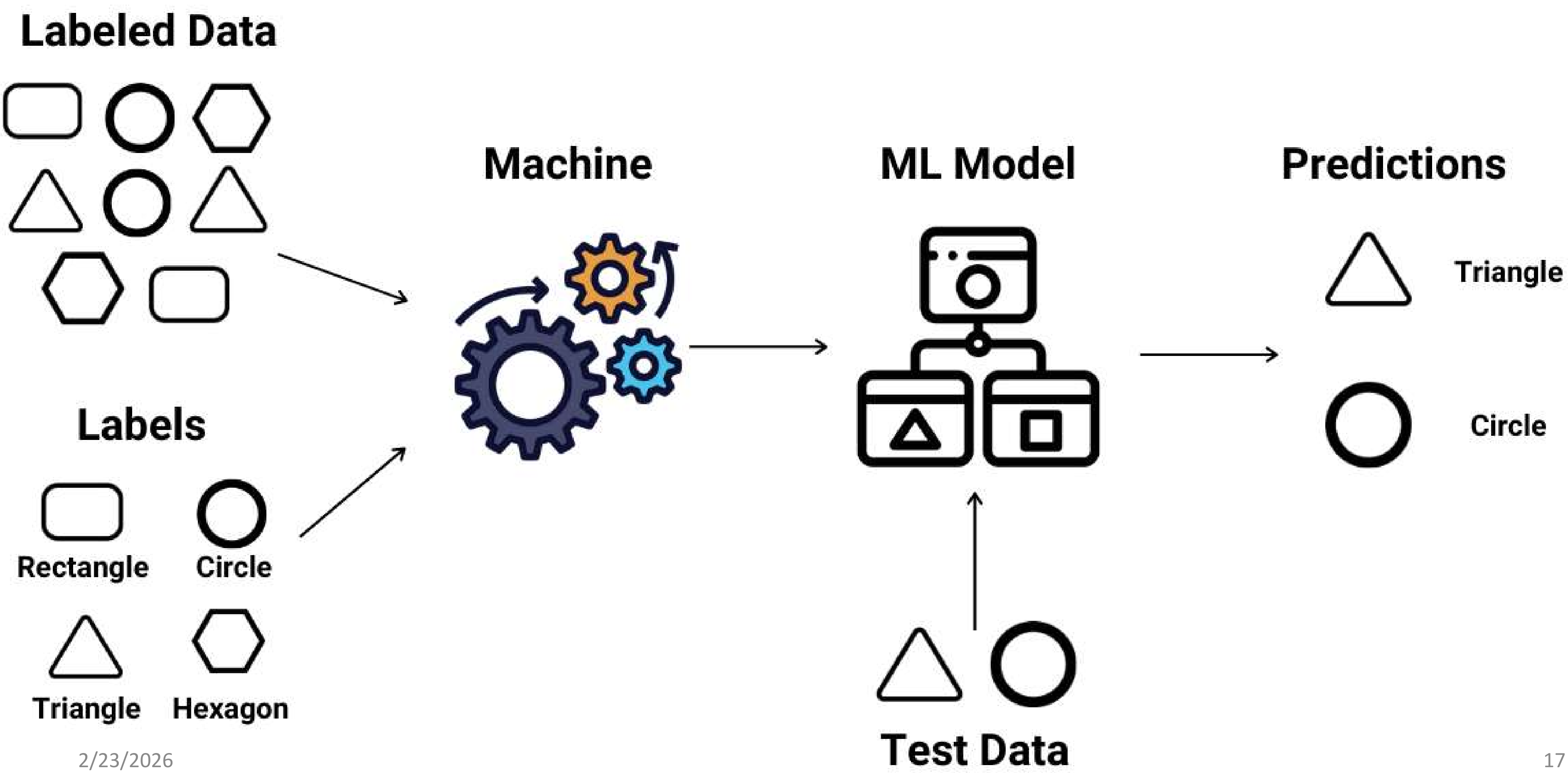
Supervised Learning

- There are two major types of supervised learning:
  - Classification:
    - A classification problem is when the output variable is a category, such as red or blue or disease and no disease.
  - Regression:
    - A regression problem is when the output variable is a real value, such as dollars or weight, age, height.
- Classification aims to predict a discrete class label. Regression, on the other hand, is the challenge of predicting a continuous variable.

- Goal: to map input data with the output data. f(x1) → yi where I is labels.
- It is based on supervision, and it is same as when a student learns things in research in the supervision of the research supervisor.



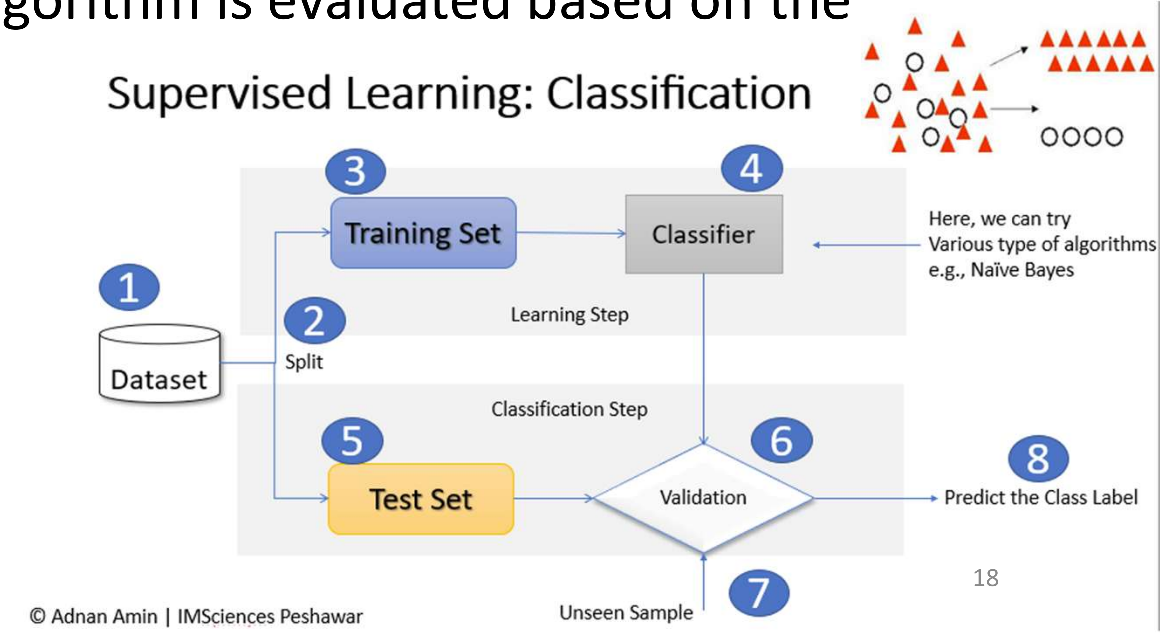
Supervised Learning



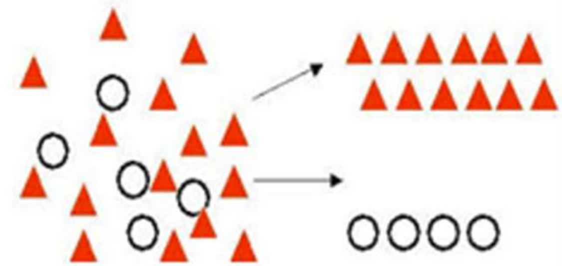
Classification

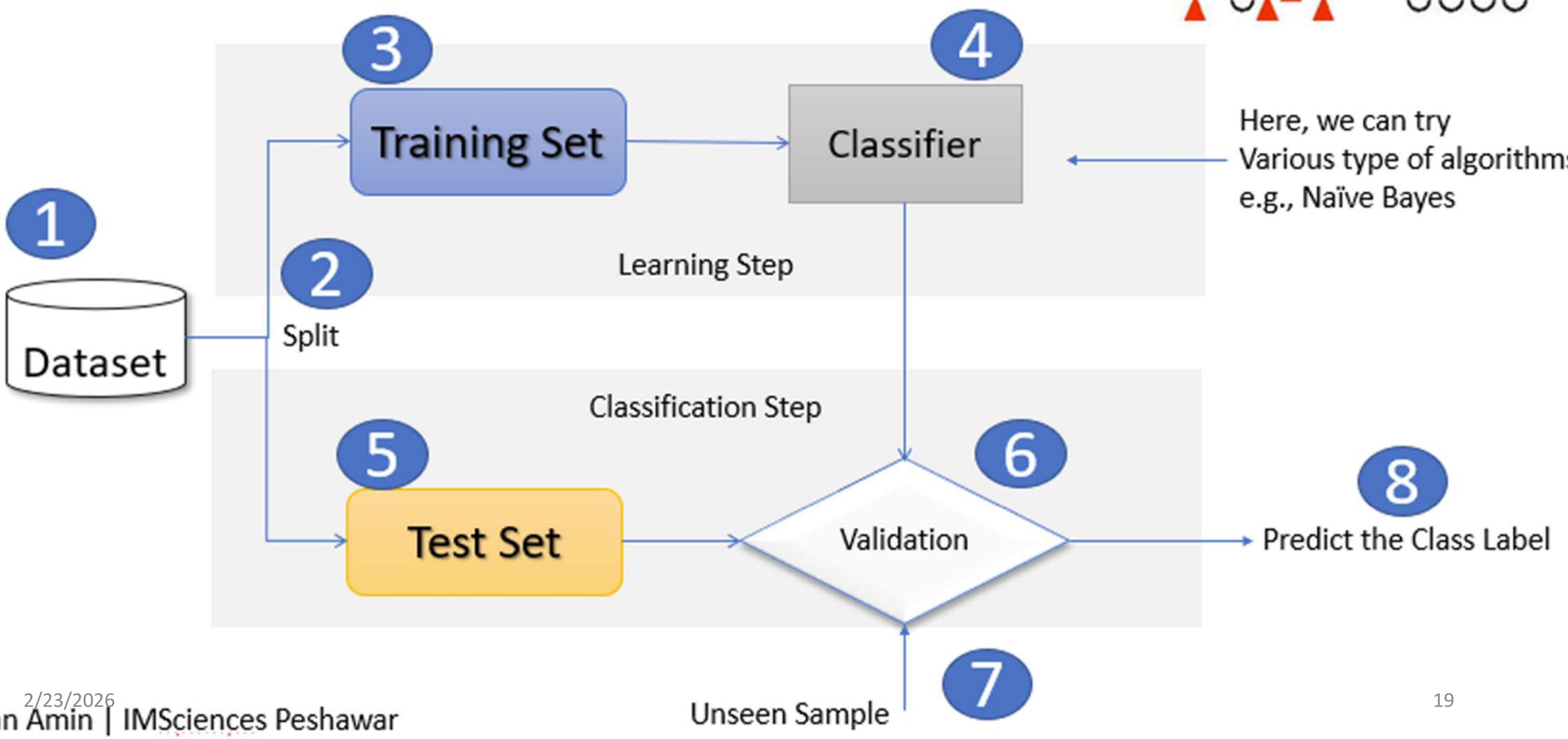
- In ML, classification refers to a predictive modeling problem in which class labels are predicted for a given example of input data.
  - E.g., Categorizing email as "spam" or "no-spam“
- The classification prediction algorithm is evaluated based on the outcomes.

We divide the dataset into training set and testing set.
Reason: we want to test the model after we are done, so a test set is used for evaluating the designed model.



Supervised Learning: Classification





1. Supervised learning (types)

- Supervised learning can be categorized into two types of algorithms:
  - Classification (output will be categorical/interval)
  - Regression (output will be continuous value)
- Well-known supervised algorithm:
  - K-nears neighbors
  - Linear regression
  - Neural networks
  - Support vector machines
  - Logistic regression
  - Decision trees and random forests

E.g. Prediction of house price

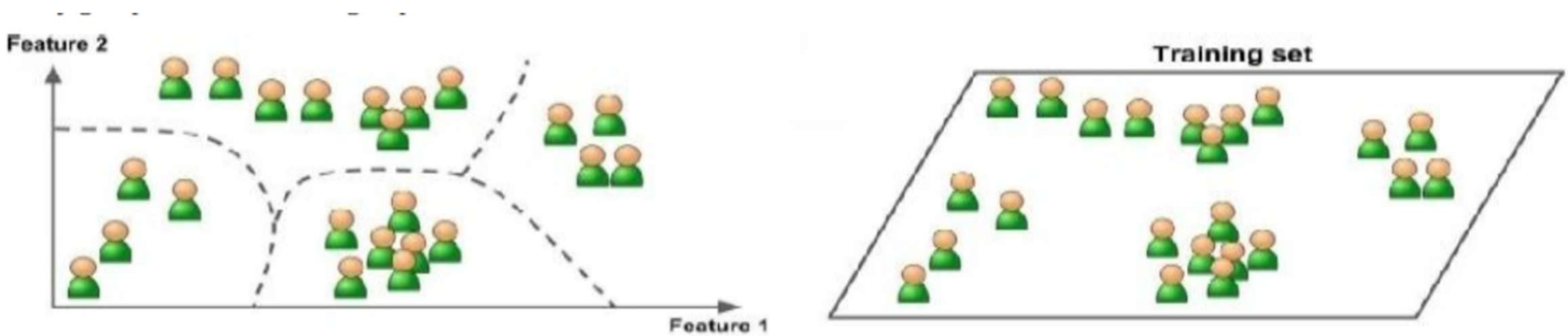
Size of the house → Size of the house

$f(x1) \rightarrow y$

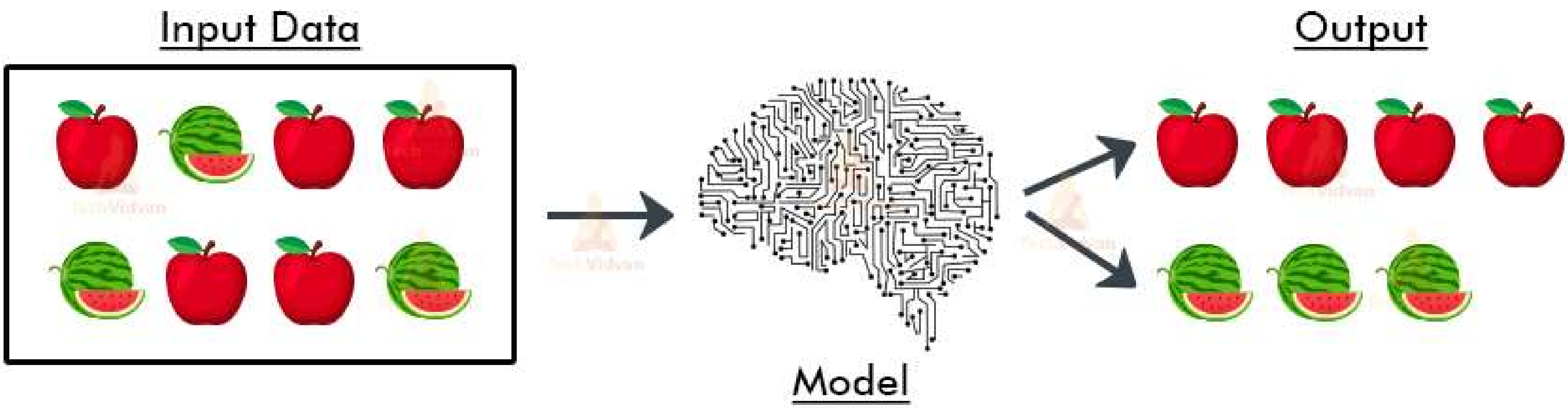
Where x1 is a feature or predictor and y is a label or target variable

2. Unsupervised Learning

- A learning method in which a machine learns without any supervision.
- You can guess that the data is unlabeled.
- Unsupervised learning sounds awful! Why use it?
  - Because you don’t know what you’re looking for:



Unsupervised Learning in ML



2. Unsupervised Learning (algorithms)

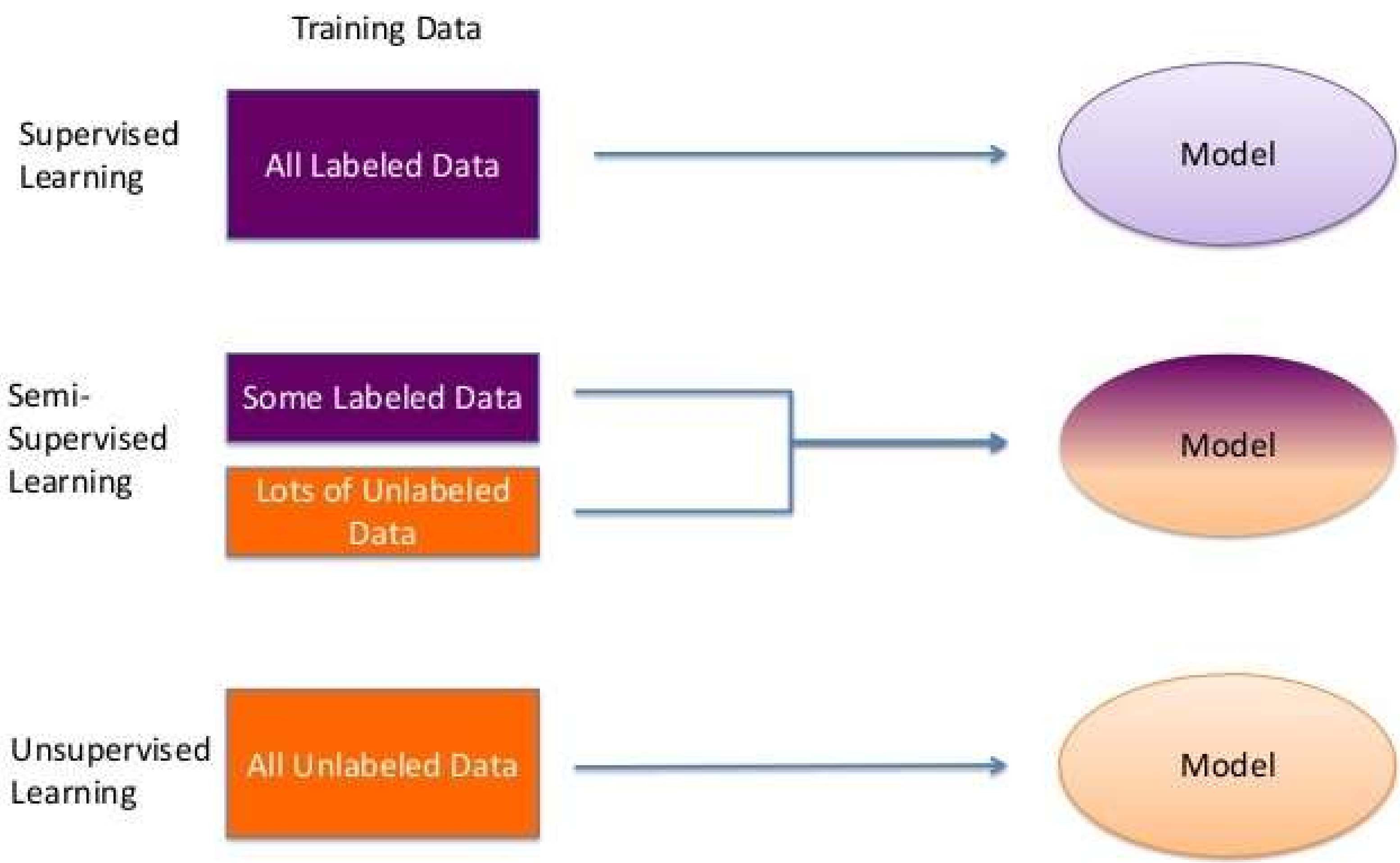
- Well-known unsupervised algorithms:
  - Clustering (k-means, hierarchical cluster analysi)
  - Association rule learning (Eclat, apriori, FP-Growth)
  - Visualization and dimensional (PCA)

Supervised vs Unsupervised

- In unsupervised learning, it is usually cheaper to collect training data because it does not require extra human efforts to label each input with the corresponding output. But much harder problem because of the lack of supervision information.
- In supervised learning, collecting the input–output pairs for supervised learning often requires human annotation, which may be expensive in practice.

3. Semi-supervised Learning

- In between these two supervised and unsupervised learning,
  - we can combine a small amount of labeled data with a large amount of unlabeled data during training.
- These learning methods are often called semi-supervised learning.
- In special case:
  - When the true outputs is difficult to or expensive to obtain, we can use other readily available information, which is only partially relevant to the true outputs, as some weak supervision signals in learning.
- These methods are called weakly supervised learning.

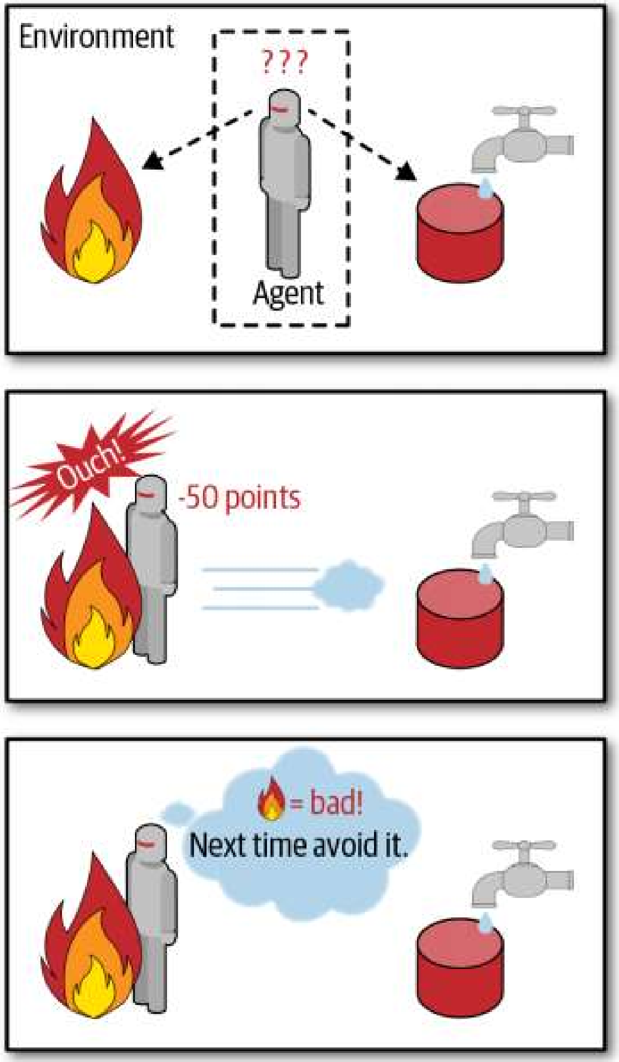

Training Data
Supervised Learning
All Labeled Data
Model
Semi-Supervised Learning
Some Labeled Data
Lots of Unlabeled Data
Model
Unsupervised Learning
All Unlabeled Data
Model

4. Reinforcement Learning

- An agent "AI System" will observe the environment, perform given actions, and then receive t rewards in return. With this type, the agent must learn by itself. Ties called a policy.

  - OR

- Reinforcement learning is a feedback-based learning method, in which a learning agent gets a reward for each right action and gets a penalty for each wrong action. The agent learns automatically with these feedbacks and improves its performance. The agent interacts with the environment and explores it. The goal of an agent is to get the most reward points, and hence, it improves its performance.



1. Observe
2. Select action using policy
3. Action!
4. Get reward or penalty
5. Update policy (learning step)
6. Iterate until an optimal policy is found

Further ML types

- Batch Learning
- Online learning
- Instance based learning
- Model-based learning

5. Self-supervised learning

- actually generating a fully labeled dataset from a fully unlabeled one
- Self-supervised learning is a type of machine learning that helps a model learn from
- unlabeled data by creating its own labels. It is a special case of supervised learning where the model generates labels automatically instead of relying on human-labeled data.

Step-by-Step Explanation: SSL

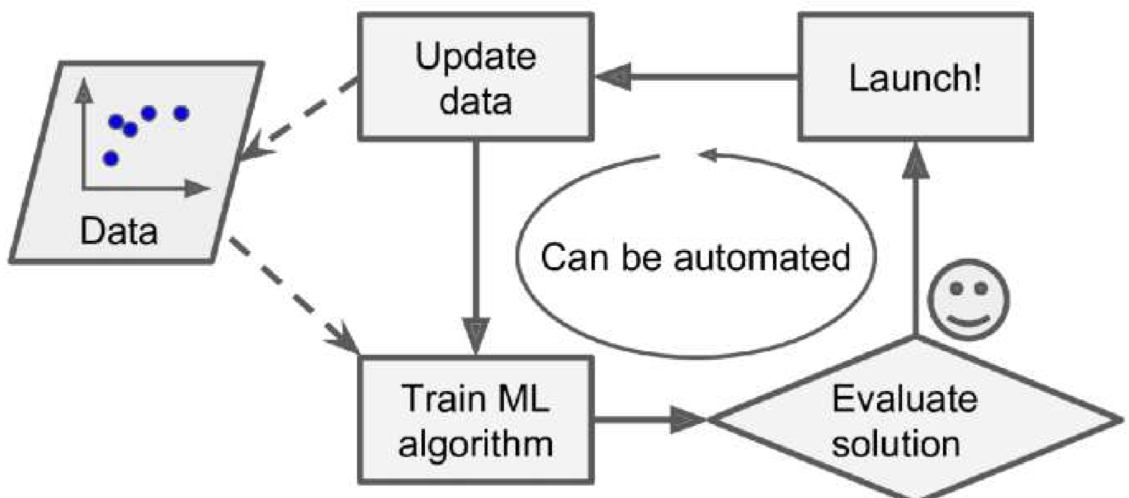
- **Hide Part of the Data and Force the Model to Guess It**
  - Suppose you have an image of a cat. You randomly mask (hide) a small part of the image.
  - The model's task is to predict the missing part based on the rest of the image.
  - If the model correctly reconstructs the missing part, it means it has learned the important patterns of what a cat looks like.
- **Learning Through Repetition**
  - The model repeats this process with thousands of images, getting better over time at predicting missing parts.
  - Eventually, it understands what features distinguish cats from dogs **without ever being explicitly told**.

Step-by-Step Explanation: SSL

- **Using the Learned Knowledge for Other Tasks**
  - Once the model has learned to understand images, we can **fine-tune** it for other tasks.
  - For example, we can now train it to classify pet species by using a small labeled dataset.
  - The model already knows what cats and dogs look like, so it only needs a little extra training to
  - map these patterns to labels like "cat" or "dog."

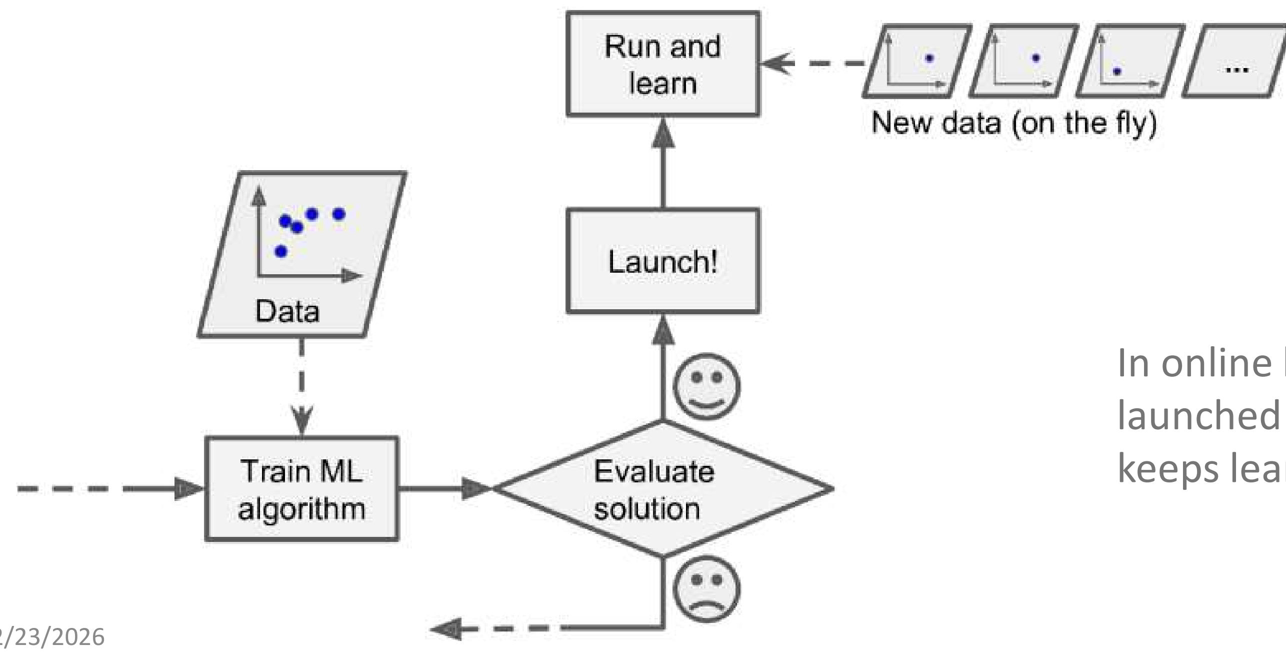
1. Batch Learning ML

- In this, the system will require prior all data before training the machine, so it’s always done offline.
- The first step to do is to train the system, and then launch it without any learning.



2. Online Learning

- The system can learn incrementally by providing the system with all the available data as instances, and then the system can learn on the fly



In online learning, a model is trained and launched into production, and then it keeps learning as new data comes in

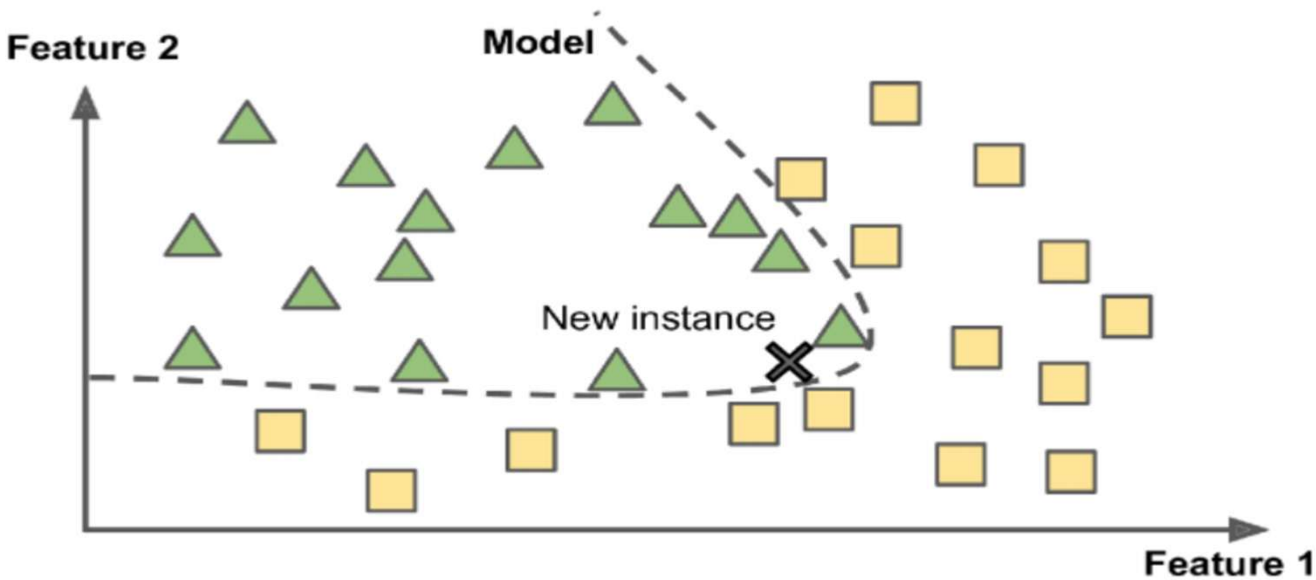
3. Instance-based learning ML

- Memory based learning
- compares new problem instances with instances seen in training, which have been stored in memory.
- Well-known Instance-based learning algorithms
  - K Nearest Neighbor (KNN)
  - Self-Organizing Map (SOM)

The new instance would be classified as a triangle because the majority (2 Triangles vs 1 Square) of the most similar instances belong to that class.

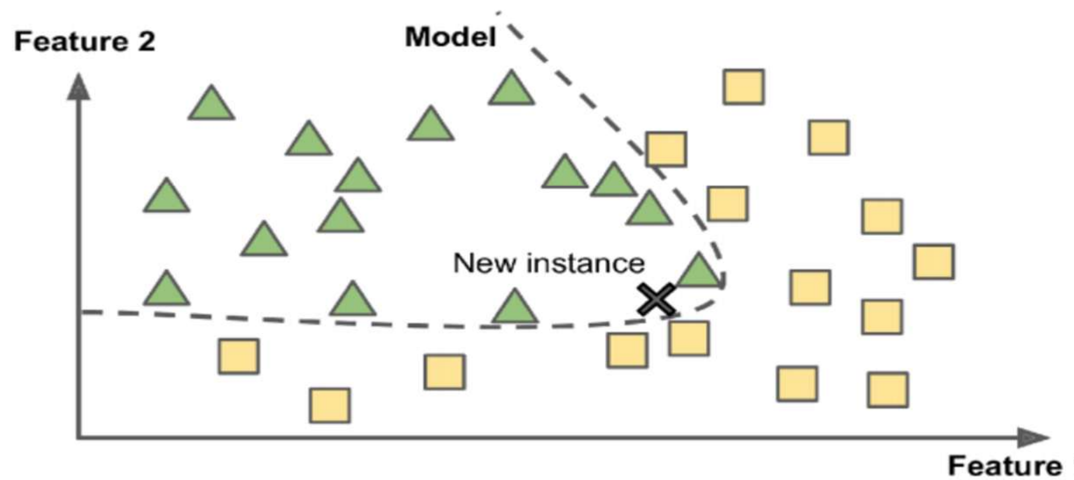
4. Model-Based Learning ML

- usually refers to a **ML** model's ability to perform well on new unseen data rather than just the data that it was trained on.



The new instance would be classified as a triangle no matter the majority of the most similar instances belong to Squar(1 Triangle vs 2 Squares) as the model defines so.

The new instance would be classified as a triangle because the majority (2 Triangles vs 1 Square) of the most similar instances belong to that class.



The new instance would be classified as a triangle no matter the majority of the most similar instances belong to Squar(1 Triangle vs 2 Squares) as the model defines so.

Machine Learning Pipeline

- Data Collection (Sensors, logs, databases)
- ↓
- Data Preprocessing (Missing values, noise)
- ↓
- Feature Engineering (Numerical representation of reality)
- ↓
- Model Selection (Mathematical function)
- ↓
- Training (Learning parameters)
- ↓
- Evaluation (Trustworthiness)
- ↓
- Deployment (Real-world usage)

Data Mining vs Machine Learning (Critical Concept)

Aspect	Data Mining	Machine Learning
Goal	Knowledge discovery	Prediction & decision making
Focus	Patterns & rules	Generalization
Output	Rules, clusters	Models
Evaluation	Interestingness	Accuracy / error
Usage	Descriptive	Predictive

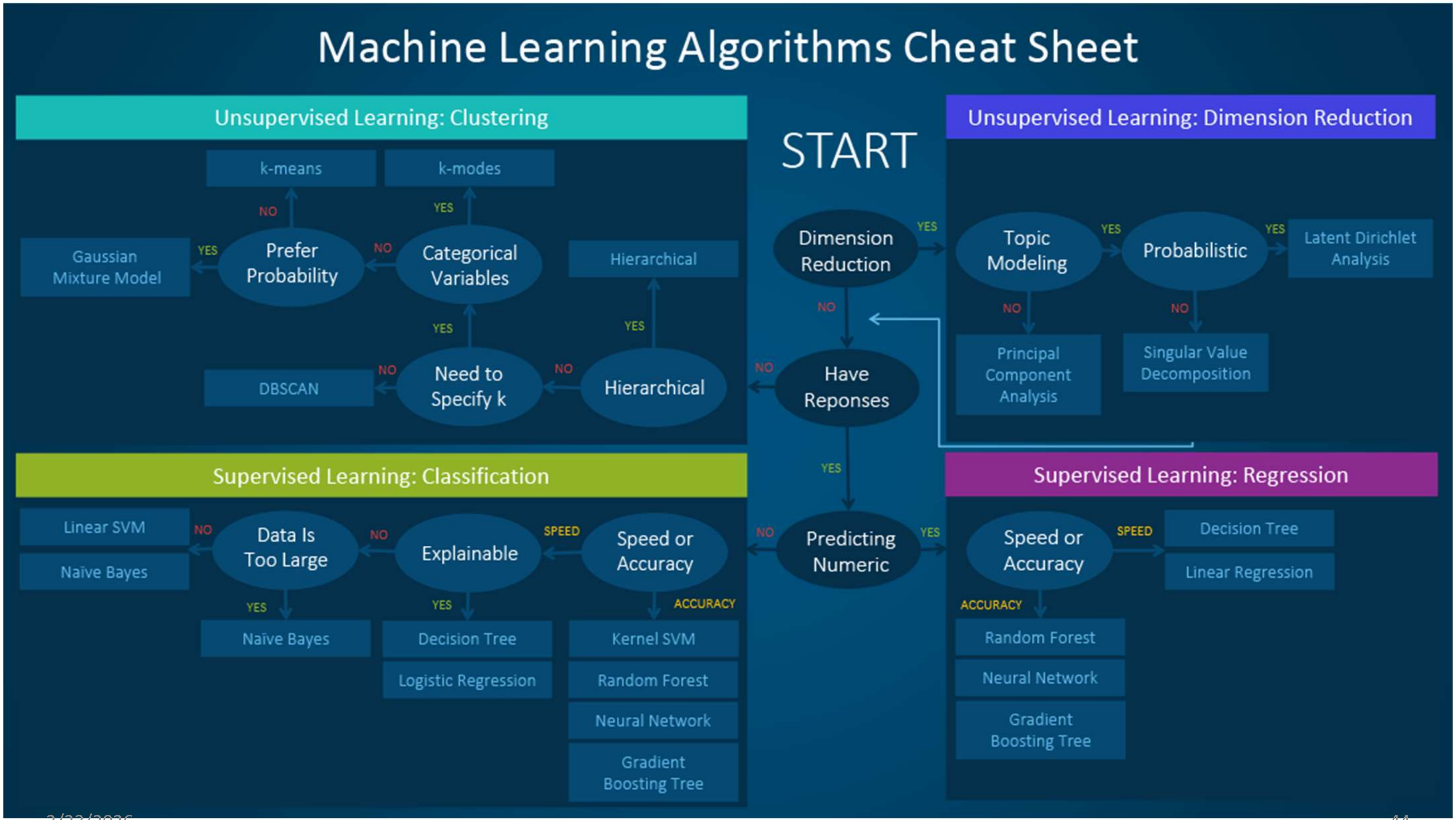
Data Mining explains the past, Machine Learning predicts the future.

Why ML when we already studied Data Mining?

- Limitations of traditional DM:
  - Rule explosion (Apriori)
  - Scalability issues
  - Poor generalization
  - Manual thresholding
- ML addresses:
  - Automation
  - Optimization
  - Continuous improvement
  - Generalization to unseen data

Main Challenges of Machine Learning

- the two things that can go wrong are “bad model” and “bad data”. Let’s start with examples of bad data.
- **Bad Data :-**
  - **1. Insufficient Quantity of Training Data**
  - **2. Nonrepresentative Training Data**
  - **3. Poor-Quality Data**
  - **4. Irrelevant Features**


Machine Learning Algorithms Cheat Sheet
START
Unsupervised Learning: Clustering
k-means
k-modes
NO
YES
Gaussian Mixture Model
YES
Prefer Probability
NO
Categorical Variables
Hierarchical
YES
YES
DBSCAN
NO
Need to Specify k
NO
Hierarchical
Dimension Reduction
YES
NO
Have Reponses
NO
YES
Unsupervised Learning: Dimension Reduction
Topic Modeling
YES
Probabilistic
YES
Latent Dirichlet Analysis
NO
NO
Principal Component Analysis
Singular Value Decomposition
Supervised Learning: Classification
Linear SVM
Naïve Bayes
NO
Data Is Too Large
NO
Explainable
SPEED
Speed or Accuracy
YES
YES
ACCURACY
Naïve Bayes
Decision Tree
Logistic Regression
Kernel SVM
Random Forest
Neural Network
Gradient Boosting Tree
NO
Predicting Numeric
YES
Supervised Learning: Regression
Speed or Accuracy
SPEED
Decision Tree
Linear Regression
ACCURACY
Random Forest
Neural Network
Gradient Boosting Tree


Machine Learning Algorithms
Deep Learning
Deep Boltzmann Machine (DBM)
Deep Belief Networks (DBN)
Convolutional Neural Network (CNN)
Stacked Auto-Encoders
Ensemble
Random Forest
Gradient Boosting Machines (GBM)
Boosting
Bootstrapped Aggregation (Bagging)
AdaBoost
Stacked Generalization (Blending)
Gradient Boosted Regression Trees (GBRT)
Neural Networks
Radial Basis Function Network (RBFN)
Perceptron
Back-Propagation
Hopfield Network
Regularization
Ridge Regression
Least Absolute Shrinkage and Selection Operator (LASSO)
Elastic Net
Least Angle Regression (LARS)
Rule System
Cubist
One Rule (OneR)
Zero Rule (ZeroR)
Repeated Incremental Pruning to Produce Error Reduction (RIPPER)
Regression
Linear Regression
Ordinary Least Squares Regression (OLSR)
Stepwise Regression
Multivariate Adaptive Regression Splines (MARS)
Locally Estimated Scatterplot Smoothing (LOESS)
Logistic Regression
Bayesian
Naive Bayes
Averaged One-Dependence Estimators (AODE)
Bayesian Belief Network (BBN)
Gaussian Naive Bayes
Multinomial Naive Bayes
Bayesian Network (BN)
Decision Tree
Classification and Regression Tree (CART)
Iterative Dichotomiser 3 (ID3)
C4.5
C5.0
Chi-squared Automatic Interaction Detection (CHAID)
Decision Stump
Conditional Decision Trees
M5
Dimensionality Reduction
Principal Component Analysis (PCA)
Partial Least Squares Regression (PLSR
Sammon Mapping
Multidimensional Scaling (MDS)
Projection Pursuit
Principal Component Regression (PCR)
Partial Least Squares Discriminant Analysis
Mixture Discriminant Analysis (MDA)
Quadratic Discriminant Analysis (QDA)
Regularized Discriminant Analysis (RDA)
Flexible Discriminant Analysis (FDA)
Linear Discriminant Analysis (LDA)
Instance Based
k-Nearest Neighbour (kNN)
Learning Vector Quantization (LVQ)
Self-Organizing Map (SOM)
Locally Weighted Learning (LWL)
Clustering
k-Means
k-Medians
Expectation Maximization
Hierarchical Clustering